

Armin Rezaiyan-Nojani

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EDUCATION

University of Maryland, College Park <i>Bachelor's of Science, Computer Science</i> - GPA: 3.3/4.0 - Clubs: Terps Racing - Relevant Coursework: Computer Systems, Computer Networks & Security, Data Structures, Algorithms, Programming Language Theory, Artificial Intelligence, Data Science, Discrete Mathematics, Statistics & Probability, Calculus I/II, Linear Algebra	College Park, MD <i>Jan 2024 — May 2026</i>
Thomas S. Wootton High School <i>High School Diploma</i> Varsity Soccer (4 years), Montgomery College Early College Program	Rockville, MD <i>Sep 2019 — May 2023</i>

WORK EXPERIENCE

Research and Development Engineer Intern Auria - Designed and deployed AI agents for satellite mission planning, enabling industry-specific workflows. - Built a data pipeline with C# and Python to extract satellite data, enabling seamless ingestion by the agent. - Led evaluation and adoption of AWS Bedrock, MCP integration, and driving production rollout. - Co-authored a SmallSat 2025 publication on closed-loop tasking workflows with conversational agents.	May 2025 — Present <i>Colorado Springs, CO</i>
Software Developer Intern Vortex Vector - Built a hardware compatibility engine to automate component matching for custom glass assemblies. - Designed a DynamoDB schema, unifying fragmented hardware catalogs across manufacturers. - Utilize Python data parsers to extract, normalize, and filter product attributes for system integration. - Accelerated design workflows by standardizing hardware selection and reducing manual lookup time for engineers.	Jun 2024 — Sep 2024 <i>Silver Spring, MD</i>
Machine Learning Intern NASA - Built a vector database and knowledge graph to improve discoverability of NASA's GES DISC archives. - Linked PDF metadata and datasets using semantic embeddings for context-aware search. - Integrated Neo4j relationships to enhance retrieval of related scientific materials. - Benchmarked and published results at AGU 2023	Jun 2023 — Aug 2023 <i>Greenbelt, MD</i>
Machine Learning Intern NASA - Built an XGBoost Learning-to-Rank model to improve dataset relevance in NASA's GES DISC search engine - Processed 230K+ user logs in Python to create training datasets and judgment lists. - Engineered ranking features, boosting nDCG from 0.36 to 0.81. - Delivered a prototype model and data pipeline adopted for future search-ranking research.	Jun 2022 — Aug 2022 <i>Greenbelt, MD</i>

PUBLICATIONS

[1] N. Dhingra, S. Kearns, N. Evans, E. Herz, and A. Rezaiyan-Nojani, "Closed-Loop Satellite Planning and Scheduling for Low-Latency Data Collection and Analysis with a Conversational Agent Interface," in *Small Satellite Conference*, Salt Lake City, UT, 2025. [Online]. Available: <https://digitalcommons.usu.edu/smallsat/2025/all2025/105/>

[2] L. Kamper-Hinson, A. Rezaiyan-Nojani, and A. Mehrabian, "Enhancing Data and Knowledge Discoverability at NASA GES-DISC through Knowledge Graphs and Language Models," in *AGU Fall Meeting Abstracts*, 2023, pp. IN54A–1. [Online]. Available: <https://ui.adsabs.harvard.edu/abs/2023AGUFMIN54A..01K/abstract>

SKILLS

- Languages: Python, C++, C#, TypeScript, SQL, HTML/CSS, Assembly (MIPS, x86)
- Full-Stack/Cloud Development: React, Node.js, Express, MongoDB, PostgreSQL, REST APIs, Docker, AWS (S3, Lambda, DynamoDB)
- AI & Data Systems: Pandas, scikit-learn, NumPy, Pytorch, Neo4j, Weaviate/Pinecone, LangChain/LangGraph, Strands, AWS Bedrock